

Efficacy and Safety of FDC of Dapagliflozin, Glimepiride and Metformin ER tablets in Indian Type 2 Diabetes: A Subgroup Analysis of Patients Aged < 50 Years

Abstract Number:
A-1-127173-RSSDI2023

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BACKGROUND AND RATIONALE

- There is an increased risk of diabetes and associated complications among Indians¹
- CAD is prevalent at a younger age in the Indian population, with > 50% of CV disease mortality affecting individuals < 50 years of age.² In addition, diabetic-chronic kidney disease occurs in 62.3% of Indians.³
- Dapagliflozin, an SGLT2 inhibitor, increases glucose excretion and significantly decreases the risk of heart failure and hospitalization for stroke in patients with T2DM and CV issues. It also reduces the risk of adverse renal outcomes.^{4,5}
- Modern sulfonylureas like Glimepiride are considered CV-neutral and offer CV safety without increased risk of CV outcomes⁶
- An FDC of Metformin and Dapagliflozin improves CV outcomes and helps achieve greater reductions in HbA1c⁴
- Glimepiride and Metformin combination can achieve good and long-term glycemic control and are CV safe in diabetic patients⁷
- A triple FDC consisting of Dapagliflozin, Glimepiride, and Metformin may provide better glycemic control, have cardio- and reno-protective effects, and improve compliance.

OBJECTIVE

To assess the efficacy and safety of FDC of Dapagliflozin, Glimepiride, and Metformin HCl ER tablets in comparison to FDC of Metformin HCl PR and Glimepiride tablets in patients with T2DM. The efficacy and safety results in a subgroup analysis of patients < 50 years of age at 16 Weeks (N = 213) are presented here.

MATERIAL & METHODS

Study design: A multicenter, randomized, active-controlled, open-label, parallel-group, two-arm, phase III study.

Primary endpoint: Mean change in HbA1c from Baseline to Week 16

Secondary endpoints: TEAEs

Inclusion criteria: T2DM patients of either gender, aged 18–65 years

- HbA1c: $\geq 8\%$ – $\leq 11\%$
- BMI: $\leq 45 \text{ kg/m}^2$
- Who, along with diet and exercise control, is on a stable total daily dose of Glimepiride 1 mg and Metformin SR/PR/ER 1000 mg for at least 8 weeks prior to screening

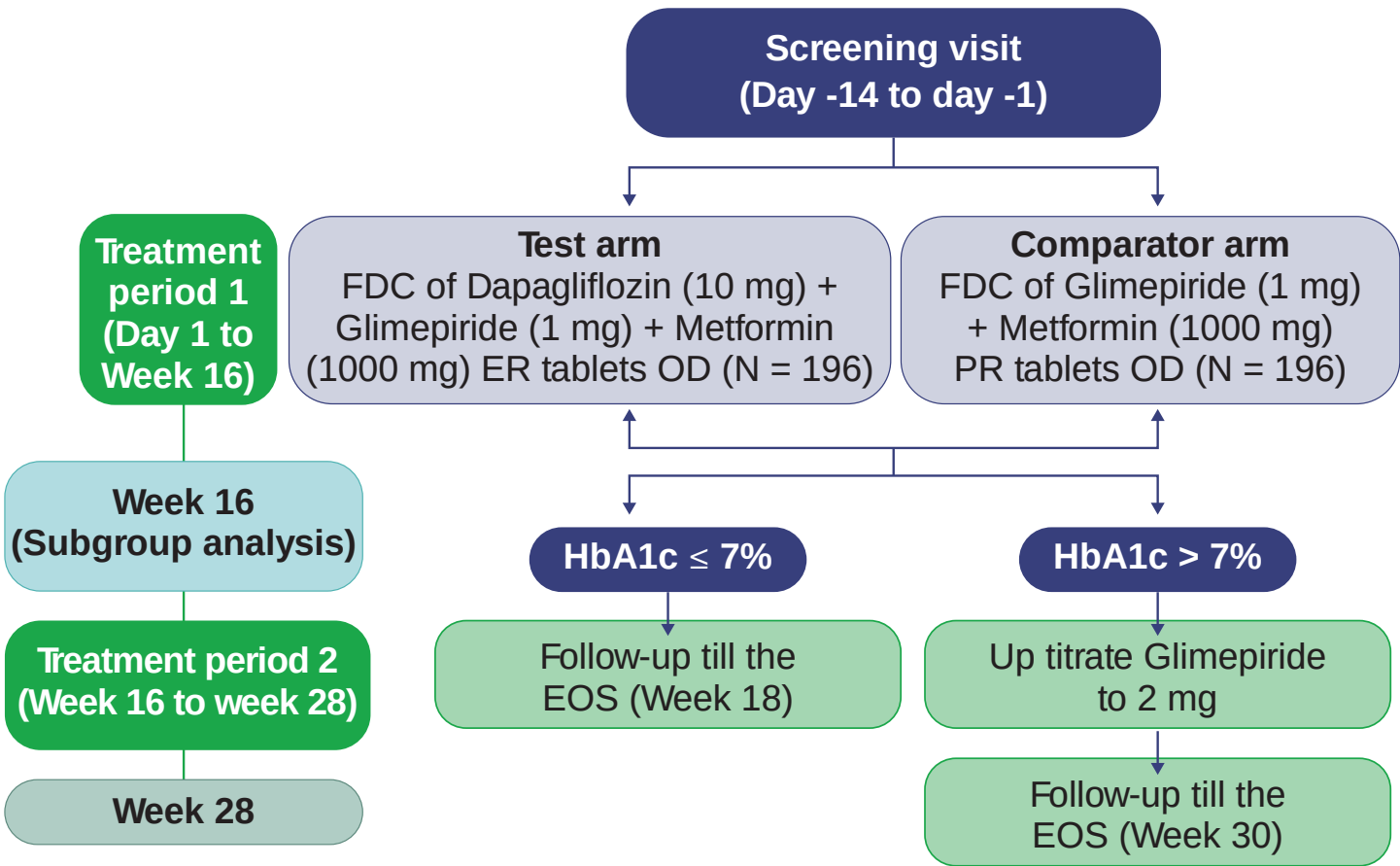
Exclusion criteria:

- FBG $\geq 270 \text{ mg/dL}$ at screening
- Any weight loss medications within 3 months prior to randomization

- Significant renal impairment (eGFR < 45 mL/min/1.73 m²) or hepatic impairment (AST and ALT > 3 times upper limit of normal)
- Uncontrolled hypertension $\geq 160/100 \text{ mmHg}$

Study arms:

- Test arm: FDC of Dapagliflozin, Glimepiride, and Metformin ER tablets (10 mg + 1 mg + 1000 mg) OD
- Comparator arm: FDC of Glimepiride + Metformin PR tablets (1 mg + 1000 mg) OD



RESULTS

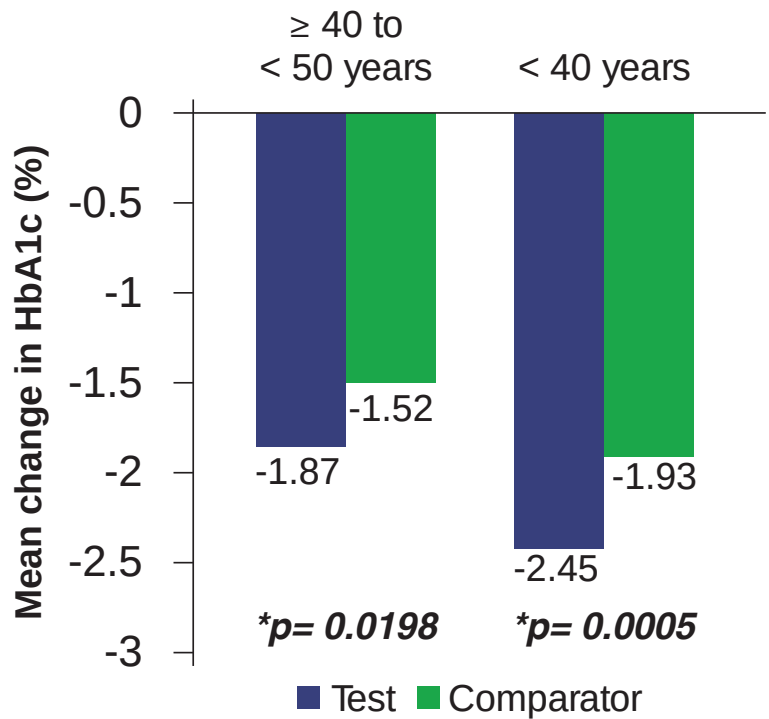
Baseline: N = 213 patients < 50 years of age
Comparable HbA1c levels in both arms (Table 1)

Table 1: Baseline HbA1c (Mean ± SD) levels in Test vs Comparator arms		
	Test	Comparator
≥ 40 to < 50 years	9.10 ± 0.89 (N = 56)	8.98 ± 0.68 (N = 62)
< 40 years	8.95 ± 0.77 (N = 45)	9.02 ± 0.71 (N = 50)

Week 16:

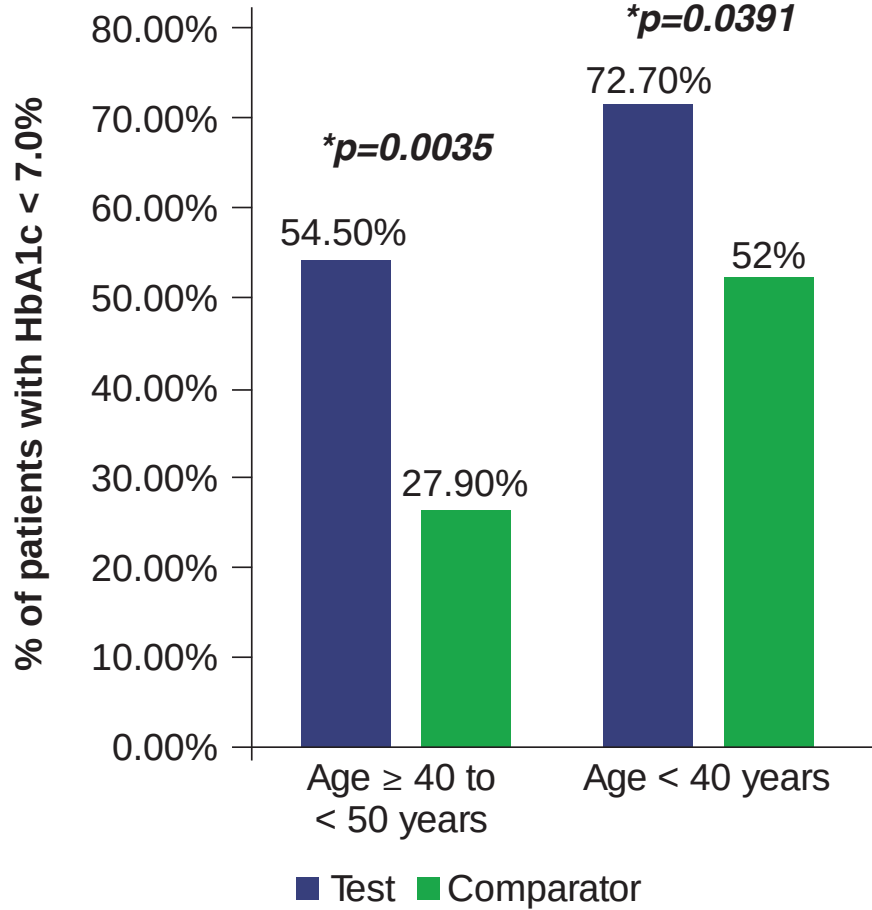
- Significant reduction in HbA1c from Baseline in all groups
- Adjusted mean change (SE) in HbA1c was significantly better in the Test group vs Comparator in participants aged ≥ 40 to < 50 years or < 40 years (Figure 1)
- Participants in the age group ≥ 40 to < 50 years or < 40 years achieving HbA1c < 7% was significantly higher in the Test arm vs Comparator arm (Figure 2)

Figure 1: HbA1c Change by Age Group at Week 16



P value vs Comparator

Figure 2: Proportion of Participants Achieving HbA1c < 7.0% at Week 16



P value vs Comparator

Safety:

- Across age groups, only 3 TEAEs were observed in both the test and comparator arms. These events were characterized as mild and did not require any treatment interruption, discontinuation, or dose reduction or lead to fatalities.
- In patients < 40 years, 1 TEAE in the test arm was likely related, while the remaining events were unlikely to be related. In the comparator arm, all TEAEs were unlikely to be related.
- In patients ≥ 40 to < 50 years, 1 TEAE in the comparator arm was likely related, while the remaining events were unlikely to be related. In the test arm, all TEAEs were unlikely to be related.
- All the medications were well-tolerated and did not cause any SAE.

CONCLUSION

- FDC of Dapagliflozin + Glimepiride + Metformin ER demonstrated significantly superior efficacy over FDC of Glimepiride + Metformin PR in reducing HbA1c from Baseline to Week 16 in T2DM patients aged < 50 years
- Study medications were well-tolerated.

References:

- Kaur KK, et al. Role of combination therapy with SGLT2 inhibitor with metformin as initial treatment for type2 diabetes-advantages of oral fixed drug pill like empagliflozin/metformin in patients with cardiovascular and renal risk-a short communication. *Archives of Diabetes and Endocrine System*. 2019;2(1):15–19.
- Ray S, et al. Usage pattern of glimepiride/metformin fixed-dose combination in type 2 diabetes patients with CVD or at risk of CVD: an experience in Indian setting. *Asian Journal of Diabetology*. 2022 Apr-Jun;23(2):13–19.
- Cahn A, et al. Cardiovascular, renal, and metabolic outcomes of dapagliflozin versus placebo in a primary cardiovascular prevention cohort: analyses from DECLARE-TIMI 58. *Diabetes Care*. 2021 May;44(5):1159–1167.
- Dalal J, et al. Vascular disease in young Indians (20–40 years): role of ischemic heart disease. *J Clin Diagn Res*. 2016 Sep;10(9):OE08–OE12.
- Hussain S, et al. Diabetic kidney disease: An overview of prevalence, risk factors, and biomarkers. *Clinical Epidemiology and Global Health*. 2021;2–6.
- Nagarathna R, et al. Prevalence of diabetes and its determinants in the young adults Indian population-call for yoga intervention. *Front Endocrinol (Lausanne)*. 2020 Dec 11;11:507064.
- Sahay RK, et al. Glimepiride and metformin combinations in diabetes comorbidities and complications: real-world evidence. *Cureus*. 2020 Sep 28;12(9):e10700.